

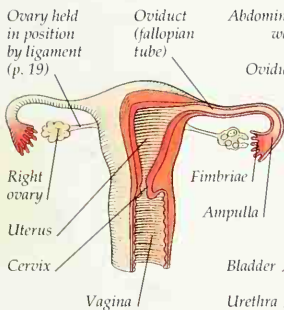


REGNIER DE GRAAF

Dutch anatomist Regnier de Graaf (1641-1673) carried out many important studies at Delft. After work on the analysis of pancreatic juices (p. 36), he published detailed works on the male and female reproductive organs in 1668 and 1672. He identified the tiny "bubbles" on the ovary's surface as ova, or eggs. Later it was discovered that each bubble is a ripe follicle, or egg-container, with the much smaller egg inside it. About 90 years later, von Haller (p. 30) named them Graafian follicles in his honor.

FEMALE REPRODUCTIVE ORGANS

In this model, the right ovary is against the abdominal wall, while the uterus is cut in section. When a ripe egg leaves its follicle it is caught by the fingerlike fimbriae and wafted into the trumpet-shaped ampulla. This leads to the curving oviduct, also named the fallopian tube after anatomist and nobleman Gabriele Fallopius of Italy (1523-1562). Fallopius published the first detailed account of this tube in 1561. The oviduct opens into the upper corner of the hollow, pear-shaped uterus. If the egg does not meet a sperm cell and become fertilized, it dies and is lost with the menstrual blood during the period.



Front view of female reproductive system

The menstrual cycle

A man's testes make sperm all the time. If not released, they die and are reabsorbed. The female system goes through hormonal changes (pp. 40-41) in a cycle lasting about four weeks – the menstrual cycle. The cycle occurs from puberty until menopause, unless pregnancy begins.

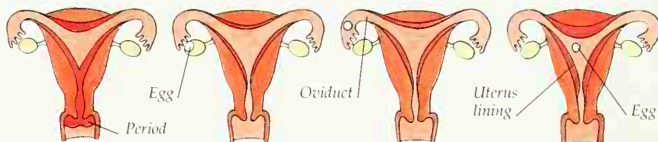


RIPE EGG

Each egg cell ripens in a fluid-filled container called a follicle, like a blister on the ovary's surface. This electron micrograph shows a just-released (ovulated) egg cell (red), ready to enter the oviduct. It has left the ripe Graafian follicle (upper left).

HORMONE CONTROL

Follicle-stimulating hormone from the pituitary gland (p. 57) ripens the egg follicle. Luteinizing hormone from the pituitary stimulates the follicle to release its egg at ovulation. The empty follicle is called a corpus luteum and makes the hormone progesterone. This, with oestrogen from the ovary, thickens the uterus lining.



1 FIRST WEEK
The thickened lining of the uterus, prepared to nourish a fertilized egg, breaks down and is lost as blood flow out of the vagina (the period).

2 SECOND WEEK
An egg ripens near the surface of one of the ovaries, to become a Graafian follicle. The uterus lining begins to grow and thicken again.

3 THIRD WEEK
The egg bursts from its container or follicle and moves along the oviduct, massaged by muscles and microscopic cilia (hairs) in its lining.

4 FOURTH WEEK
The egg reaches the uterus. It has not joined with a sperm, so the blood-rich uterus lining is not needed, and the cycle starts over again.

